INCH-POUND

MIL-DTL-45952A
11 August 2014
SUPERSEDING
MIL-P-45952
5 March 1973

DETAIL SPECIFICATION

PIN, QUICK-RELEASE, DETENT GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for detent, quick-release pins (see 6.1).

1.2 Classification. Quick-release pins shall be of the style and sizes specified on the specification sheet, MIL-P-45952/1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-P-45952/1	Pin, Quick-Release, Detent

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

AMSC N/A FSC 5315

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME-B46.1	Surface Texture(Surface Roughness, Waviness, and Lay)

(Copies of this document are available from www.asme.org American Society of Mechanical Engineers, Three Park Avenue, M/S 10E, New York, NY 10016-5990.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of this document are available from www.asq.org American Society for Quality Control, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A108	Standard Specification for Steel Bar, Carbon and Alloy, Cold Finished
ASTM A228/A228M	Standard Specification for Steel Wire, Music Spring Quality
ASTM A313/A313M	Standard Specification for Stainless Steel Spring Wire
ASTM A484/A484M	General Specification for Stainless Steel Bars, Billets, and Forgings
ASTM A580/A580M	Standard Specification for Stainless Steel Wire
ASTM A582/A582M	Standard Specification for Free-Machining Stainless Steel Bars
ASTM F1941	Standard Specification for Electrodeposited Coatings on Threaded Fasteners (Unified Inch Screw Threads (UN/UNR))

(Copies of these documents are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

NATIONAL AEROSPACE STANDARD (NAS)

NASM1312-13	Fastener Test Methods Method 13 Double Shear Test

(Copies of this document are available from www.aia-aerospace.org or the Aerospace Industries Association, 1250 Eye Street, N. W., Suite 1200, Washington DC, 20005-3924.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AIR4127	Steel: Chemical Composition and Hardenability
SAE AMS2700	Passivation of Corrosion Resistant Steels
SAE AMS-C-81562	Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited)
SAE AMS-H-6875	Heat Treatment of Steel Raw Materials
SAE AMS-QQ-P-416	Plating, Cadmium (Electrodeposited)
SAE AMS-QQ-S-763	Steel, Corrosion Resistant, Bars, Wire, Shapes and Forgings
SAE AMS-S-6758	Steel, Chrome-Molybdenum (4130) Bars and Reforging Stock (Aircraft Quality)

(Copies of these documents are available from www.sae.org or the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification Sheet. The individual item requirements shall be as specified herein and in accordance with the specification sheet, MIL-P-45952/1.

3.2 Material. Pins shall be made from the materials in 3.2.1 or 3.2.2, as specified in MIL-P-45952/1.

3.2.1 Alloy steel.

3.2.1.1 Head and shank. When using one-piece construction, the head and shank shall be 4130 alloy steel in accordance with SAE AMS-S-6758. The shank shall be heat treated in accordance with SAE AMS-H-6875 to meet the mechanical properties specified in3.3. When using two-piece construction, the shank shall be 4130 alloy steel as specified above and the head shall be carbon steel in accordance with ASTM A108.

3.2.1.2 Balls. The balls shall be 440C corrosion-resisting steel in accordance with SAE AMS-QQ-S-763.

3.2.1.3 Spring and ring. The spring and ring shall be carbon steel music wire in accordance with ASTM A228/A228M.

3.2.2 Corrosion-resisting steel.

3.2.2.1 Head and shank. The head and shank shall be corrosion-resisting steel, 300 Series, in accordance with SAE AMS-QQ-S-763, ASTM A484/A484M or ASTM A582/A582M.

3.2.2.2 Balls. The balls shall be 440C corrosion-resisting steel in accordance with SAE AMS-QQ-S-763.

3.2.2.3 Spring and ring. The spring and ring shall be corrosion-resisting steel wire, 300 Series, in accordance with ASTM A313/A313M or ASTM A580/A580M.

3.3 Mechanical Properties.

3.3.1 Tensile Strength. Alloy steel shanks shall be heat treated to 160,000-180,000 PSI tensile strength.

3.3.2 Double-shear strength. Pins shall be capable of withstanding, without failure, the double-shear strengths specified in MIL-P-45952/1.

3.4 Protective finish.

3.4.1 Cadmium. Alloy and carbon steel parts shall be cadmium plated in accordance with SAE AMS-QQ-P-416, Type II, Class 2 or cadmium coated in accordance with SAE AMS-C-81562, Type II, Class 2.

3.4.2 Passivation. Corrosion-resisting steel parts shall be passivated in accordance with SAE AMS2700.

3.4.3 Zinc Nickel plating. As an alternative to Cadmium plating, carbon steel inserts may be ZnNi plated in accordance with ASTM F1941 Fe/Zn-Ni 8ET alkaline zinc nickel electroplate, 12%-16% mass percent nickel, with chemical conversion coating per MIL-DTL-5541 TYPE II CLASS 1A.

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3.5 Design and construction. The design and construction of the pins shall be as specified herein and in accordance with MIL-P-45952/1.

3.5.1 Shank straightness. Pin shank straightness shall be as specified in Table I.

TABLE I. Shank Straightness.

Shank Diameter (Inch)	Maximum Deviation (Inch per Inch Grip Length)
.250	.0030
.375	.0025
.500 and up	.0020

3.5.2 Surface roughness. Pins shall have a maximum surface roughness(arithmetical average), after application of protective finish, of 125 microinches. Values shall be determined in accordance with ASME B46.1.

3.5.3 Insertion and removal force. Insertion and removal forces for these pins shall fall within the limits specified in MIL-P-45952/1 when inserted and removed 15 times.

3.5.4 Ring. The ring shall be located within a hole through the head and shall be able to rotate through the hole. The ring shall not break or deform when subjected to the maximum removal force specified in MIL-P-45952/1.

3.6 Workmanship. Pins shall be free from pits, seams, cracks, burrs, scratches and other surface contamination. Superficial scratches to the chromate coating in the staked ball area and the head-shank juncture due to fabrication techniques shall be allowed.

3.7 Material certificate. The contractor shall furnish the Government with a mill certificate covering the chemical requirements of the material being used to make the pins.

3.8 Marking. The pin shall be marked with the manufacturer's identification. Markings may be raised or depressed.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the following.

4.2 Inspection lot. A lot shall consist of completed pins which are of the same part number, produced by the same manufacturer under essentially the same conditions, and submitted for acceptance at one time.

4.3 Sampling for lot inspection.

4.3.1 Sampling for examination. A random sample of pins shall be taken from each lot in accordance with ASQ Z1.4, Inspection Level I.

4.3.2 Sampling for test of double-shear strength and insertion and removal force. A random sample of pins shall be taken from each lot in accordance with ASQ Z1.4, Inspection Level S-2.

4.3.3 Sampling for protective finish. Sampling for test of protective finish shall be in accordance with the applicable specification of 3.4.

4.4 Examination. Each pin sampled as specified in 4.3.1 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table II. Any pin containing a defect shall be rejected and if the number of defective pins in any individual category exceeds the acceptance number for that category, the lot represented by the sample shall be rejected.

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TABLE II. Classification of Defects.

Category	Defect	Inspection Method
Critical	None defined	
Major		
101	Material, not as specified (3.2)	Visual
102	Grip length, not as specified (3.5)	*SIE
103	Shank diameter, not as specified (3.5)	SIE
104	Shank straightness, not as specified (3.5.1)	SIE
Minor		
201	Protective finish, missing or incomplete (3.4)	Visual
202	Other dimensions, not as specified (3.5)	SIE
203	Surface roughness, not as specified (3.5.2)	SIE
204	Workmanship (3.6)	Visual
205	Marking (3.8)	Visual

*SIE - Standard Inspection Equipment

4.5 Tests.

4.5.1 Double-shear. Samples taken as specified in 4.3.2 shall be subjected to a double-shear test to determine conformance with 3.3.2. The test shall be conducted in accordance with NASM1312-13.

4.5.2 Insertion and removal force. Samples taken as specified in 4.3.2 shall be subjected to an insertion and removal force test to determine conformance with 3.5.3. Pins shall be inserted and removed 15 times using the correct size holes specified in MIL-P-45952/1. Maximum and minimum forces shall be within the limits specified. The ring shall not break or deform (see 3.5.4).

4.5.3 Protective finish. Tests of protective finishes shall be in accordance with the applicable specification of 3.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Detent quick-release pins are intended for use for fastening where quick assembly and disassembly are required. These pins can be inserted and removed by hand. They should be used for shear applications only.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of the specification sheet, MIL-P-45952/1.
c. Applicable part number.
d. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Cadmium
Passivation
Shank straightness

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AR
Navy - OS
Air Force - 11

Review Activity:
Army - AT, AV, CR4, GL
Navy - MC
Air Force - 99

Preparing Activity:

DLA - IS

(Project 5315-2014-002)

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